

Employment

Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Shipped 10+ production backend features end-to-end in TypeScript and Python: multi-model AI pipelines (Claude, GPT-4o, Gemini), REST APIs, event-driven workflows, and Auth0 RBAC; 923 commits via GitHub Actions CI/CD across 7 production deployments. - Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Inventory Reservation Service	Java 21 + Spring Boot + Maven	2026
- Built transactional inventory reservation API in Java 21 + Spring Boot (Maven) preventing overselling via PostgreSQL row-level locking across multi-SKU reservations; idempotent endpoints with RFC 9457 error responses and automated expiry via scheduled cleanup job. - Verified concurrency correctness with Testcontainers; Flyway, Spring Security RBAC, multi-stage Docker build, GitHub Actions CI.		
VectorLift	PyTorch ML Pipeline & Semantic Search	2026
- Built production semantic search on MS MARCO: Elasticsearch BM25 baseline, sentence-transformers bi-encoder fine-tuned in PyTorch with in-batch negatives, and MiniLM cross-encoder re-ranker; benchmarked 6 pipeline configurations with NDCG@10, MRR@10, and MAP. - Designed Kafka-driven indexing pipeline into Elasticsearch and Qdrant; FastAPI search with per-stage latency tracking.		
Demand Paging	Systems Programming in C — Xinu OS	2024
- Engineered 8-store virtual memory system in C for Xinu OS with Second-Chance page replacement; implemented inverted page tables and custom demand-paging fault handler, reducing page faults by 25% and improving memory throughput by 15%. - Added 7+ kernel system calls for memory management; debugged memory corruption at register level with GDB on bare-metal.		

Languages and Technologies

- Languages:** Java, Python, C, TypeScript, JavaScript, SQL, Bash
- AI & ML:** PyTorch, TensorFlow, sentence-transformers, LLM fine-tuning, RAG, OpenAI/Anthropic/Gemini APIs
- Backend & Data:** Spring Boot, Maven, Node.js, FastAPI, PostgreSQL, Redis, MongoDB, Kafka, Elasticsearch
- DevOps & Systems:** Docker, Kubernetes, Linux, GitHub Actions, CI/CD, Terraform, GCP, AWS

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 complete MVPs within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.